



BUILD BLOCK BRIEF · FALL 2026-27

Systems Build STUDIO LANE

Planning brief session 4 · build from session 8 · 44 students · 9 teams · courses: GCCO432, GCDS433, GCDS435, GCLA431, GCAG311, GCDD323, GCMC325, GCPF312, GCDS436, GCPF317

What your team is building

A working system built from first principles — hardware logic, data structures, algorithms, mathematics — that solves a measured public problem; the working system and its evaluation go to the authority.

ONE PROJECT, SEVERAL GRADEBOOKS

You submit **one** piece of work, and each course that marks it looks at a different dimension, out of 20. **At most eight courses mark one team file.** A course you take that falls outside those eight sets you a **lens note** instead — an individual piece on the same project from that course's angle, 3,000 words, same rubric, same 20 marks. No student carries more than two: from a third, that course marks your viva out of 30 instead.

YOUR MARKS FROM THIS PROJECT

30 the project — a team mark of 20 from each course that marks your file, plus your own viva of 10, held in class in session 14

30 the mid-term

40 the final

The three quizzes are practice and carry no marks.

This replaces coursework, not classes.

What each course marks

CODE	COURSE	WHAT THIS COURSE MARKS — 20 MARKS
GCCO432	Computer Organization & Assemble Language Ume Farwa	This course marks the machine-level account of the team's program: the disassembly read and annotated, the register and stack usage explained, the memory-access pattern measured, and the gap between the predicted cost and the time the machine actually took. 20 marks, team.
GCDS433	Discrete Structures Haris Yaseen	This course marks the proof: the loop invariant and the induction that establishes correctness, the recurrence set up for the chosen algorithm and its solution, the counting argument behind the search space, and whether the bound claimed is the bound proved. 20 marks, team.
GCDS435	Data Structures Dr Nasreen	This course marks the implementation: the structure chosen and the case made for it, the correctness of the operations under edge cases, the measured time and space against the predicted bound, and the quality of the test set the team wrote. 20 marks, team.



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CODE	COURSE	WHAT THIS COURSE MARKS — 20 MARKS
GCLA431	Linear Algebra Zaeem Farooqui	This course marks the mathematics: the matrix model set up for the team's problem, the method chosen to solve it, the account of conditioning and numerical error, and whether the results the program produces are the results the model predicts. 20 marks, team.
GCAG311	Calculus & Analytic Geometry Zaeem Farooqui	This course marks the continuous mathematics: whether the model is set up correctly, whether the limits, derivatives and integrals are right, whether the geometry is represented properly in coordinates, and whether the team can justify every step by hand. 20 marks, team.
GCDD323	Digital Logic Design Adeel Karim	This course marks the hardware: the truth table and state diagram derived from the model, the minimisation and the case made for it, the Verilog implementation and its testbench, and the timing analysis that shows the circuit meets its clock. 20 marks, team.
GCMC325	Multivariable Calculus Asif Rehman	This course marks the continuous model: the function set up for the team's problem, the derivatives and integrals computed from it, the optimisation carried out and its constraints, and whether the numbers the model produces are defensible. 20 marks, team.
GCPF312	Programming Fundamentals Yahya Rashidi	This course marks the code itself: whether the program the team ships actually runs, whether the logic is correct at the boundaries, whether the structure is readable by someone who did not write it, and whether the student can explain any line of it. 20 marks, team.

What to hand in

- **GCCO432 Computer Organization & Assemble Language** — By session 12, an annotated machine-level report on the team's program — disassembly of the hot function, a hand-written assembly version of it, cache and timing measurements — submitted with AI declaration.
- **GCDS433 Discrete Structures** — By session 13, a written proof document for the team's program — invariants, an induction for correctness, the recurrence and its solution, and the asymptotic bound — submitted with AI declaration.
- **GCDS435 Data Structures** — By session 12, a working program that solves the cluster problem four ways behind a common interface, with a benchmark harness and timing results, the test suite, and AI declaration.



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- **GCLA431 Linear Algebra** — By session 13, the linear-algebra note for the team's program — the matrix formulation, the solution method with its cost, an error and conditioning analysis, and verification against a reference computation — with AI declaration.
- **GCAG311 Calculus & Analytic Geometry** — A worked bench file by session 13 — the problem stated, the calculus solved by hand, the geometry drawn, the numerical check attached and the hand-off note to the Digital Logic team — submitted together with AI declaration.
- **GCDD323 Digital Logic Design** — By session 12, a working Verilog design of the circuit with a self-checking testbench, simulation waveforms, a gate-count and delay summary, and an FPGA demonstration where hardware allows — with AI declaration.
- **GCMC325 Multivariable Calculus** — By session 13, the modelling note for the cluster problem — the function and its domain, the gradient and Hessian analysis, the constrained optimisation, one integral quantity computed, and a sensitivity check — with AI declaration.
- **GCPF312 Programming Fundamentals** — By session 12 the team files the working program with its source, a short read-me saying how to run it, the test cases it passes and the ones it fails, and a note on what was written by hand and what was generated.

How big the file is

The written file is **3,000 words for every member**, plus or minus ten per cent — 15,000 words for a team of five, 12,000 for four, 9,000 for three — plus the artefacts: the evidence or data log with its instrument, interview notes and consent log, any program with its tests, the pitch deck, the AI declaration and the draft-to-final revision log. In the venture and client lanes the live thing itself is the largest artefact. Each member owns one section of 3,000 words with their name on it, one artefact they built, and one part they can defend alone at viva. A lens note is 3,000 words on the same project. The full format is in "The Build Block File — required format" in the kit.



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Dates

Session 4	28 Sep	Planning brief issued — twenty minutes of class: the brief, the Ideas Book and the format sheet. Teams locked since session 3. Choose the project, design the instrument or find the client, split the sections.
Guest session	Session 4, 5 or 6	A practitioner from outside the university speaks to your school, on whichever of those three sessions the speaker can make. Attendance is compulsory. Each student writes one page on it for their own file; it is not marked.
Session 7	19 Oct	One-page plan uploaded to the portal: the question, the instrument or the client charter, who owns what. Not marked — it is the gate. This week the convener merges the marking courses' requirements into the file's sections, so the team writes one document and not eight. Clients matched to teams.
Session 8	2 Nov	Build starts. Rubric issued, roles confirmed, team charter uploaded. Build-with-AI workshop for venture and client teams.
Session 9	9 Nov	Scope agreed with the convener. Client or subject confirmed. Your school arranges its industrial visit for some point between now and session 12.
Session 11	23 Nov	Draft review, inside a normal class: each course teacher sees the teams in their own section for about ten minutes each — show what exists, not what you plan. The convener samples three files per block.
Session 12	30 Nov	The artefact goes live or reaches its first complete state.
Session 13	7 Dec	Final file uploaded with its AI declaration. Venture teams evidence the first paying customer. Five of the twenty come off for each week the file is late, in every course marking it, and nothing is accepted after the Karachi Review.
Session 14	14-19 Dec	Vivas in class: each course holds its own, team by team, in its session-14 class — about three minutes per student, on your own part as that course sees it.
The Karachi Review	Tue 15 Dec	Your team exhibits at its own table, across the campus, all day. The day opens with a thirty-minute handover of studies to the public bodies who asked for them; press and industry attend. Ten panels — a teacher and an alumnus each — circulate the exhibition and score you at your table, about eight minutes a team. Not marked separately; the panel sheet goes to every course marking the file.
Weekend	—	Weekend cohorts hold all of the above in their Saturday and Sunday slots; the Karachi Review runs on a weekend afternoon.

Rules for this lane



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What counts as done

A finished piece of work — not a plan for one, not a prototype that only runs on your machine — with the process behind it documented.

Craft

The standard is what a competent professional would accept, at the level of your year. Unfinished work presented as finished is marked as unfinished.

Sources and rights

Anything you did not make yourself is credited. You may not use material you have no right to use, and asset licences are recorded in the file.

If your work involves people

Consenting adults only, with an information sheet, a consent form and a debrief. Mock cases where the subject is clinical. No real patient, client or examinee material, ever, and none of it into an AI tool.

Marks

Craft carries the weight, but the reasoning behind your choices carries the rest. Be able to say why you did it that way.

Using AI

You are expected to use Claude and similar tools — to research, to build, to draft, to model numbers. Three rules. Check everything a tool gives you before you use it, and record the check. Declare what you used in the AI declaration bound into the file, as the HEC *Framework on the Use of Generative AI Tools* (2026) requires. Be able to explain, at the viva, everything your team built. **Work you cannot explain is not credited to you.** Undeclared use, or use to fabricate data, references or results, is academic misconduct. Never put real client, patient or examinee data, or anything identifying a real person, into a tool.

BLOCK CONVENER

PROGRAMME HEAD

DATE ISSUED